James Akl

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Expertise in **robotics, AI**, and **autonomous systems**. Strong foundations in **engineering, mathematics, software, algorithms**, and **system design**. Five years of experience in **R&D, rapid prototyping, technology demonstration**, and **project leadership**.

EXPERIENCE

The AI Institute · Cambridge, MA

Research Scientist / Roboticist (2024/10 – Present)

- Conducting **research and development** in **robotics, AI, computer vision, learning, perception**, and **manipulation**.
- Working in a robotics development team on designing and implementing **more general** algorithms and methods.

Amazon · Innovation Lab (Fulfillment Technologies & Robotics)

Postdoctoral Scientist / Project Lead (2024/03 – 2024/10)

- Conducted **robotics/AI research** spanning **robot learning, perception, control, manipulation, human–robot interfacing**.
- Lead a prototyping team and developed innovative **robotic fulfillment systems**.

Applied Scientist (2023/08 – 2024/03)

- Conducted **robotics R&D** to design/implement algorithms for autonomous fulfillment and logistics.
- Developed **robotic manipulation** capabilities using **3D perception, deep learning**, task and motion **planning**, and **control**.

Worcester Polytechnic Institute · Robotics Engineering Department

PhD Candidate (2019/08 – 2023/08) with the Manipulation & Environmental Robotics Lab

- Solved challenging technical problems by developing **innovative solutions** and **novel algorithms**.
- Lead **research projects**, supervised MS students, and collaborated in multidisciplinary teams.
- Developed **software** for prototypes, simulations, and experiments (C++, Python, ROS, Bash, Linux, Git, PCL, OpenCV, …).
- Engineered **autonomous systems** for industrial applications using **robot vision, control, planning**, and **machine learning**.

Research Mentor (2022/05 – 2022/07) with the NSF REM Program

- Mentored students in **research methodology, data collection/processing**, image processing, and **hands-on experiments**.

Teaching Assistant (2019/08 – 2020/05)

- Assisted in **delivery, labs**, and **grading** of robotics courses ('Actuation', 'Sensing', 'Introduction to Robotics').

EDUCATION

PhD Robotics Engineering — Worcester Polytechnic Institute — **2019–2023**
Focus: Perception, control, manipulation · Dissertation: "Robot autonomy for scrap cutting in metal recycling"

Bach Mechanical Engineering — Lebanese American University — **2015–2019**
Minor in Mathematics with emphasis in real analysis, convex optimization, and numerical linear algebra

SKILLS

Software	C++ · Python · Bash · Git · Linux · CMake · ROS2 · OpenCV · Open3D · PyTorch · SciPy · scikit-learn
Platforms	Franka Emika · Universal Robots · BD Spot · Nvidia Jetson · Intel RealSense · Arduino · Raspberry Pi
Languages	English (native fluency) · French (professional fluency) · Arabic (professional fluency)

SERVICE

Reviewer · IEEE, Sage Publications, and IFRR

2021/06 – Present: Conducting scholarly reviews for journals (IJRR, RA-L, T-ASE, TMECH) and conferences (ICRA, IROS, CDC, Humanoids, CASE, ISRR, SSRR) in robotics, AI/ML, control, automation, mechatronics, and autonomous systems.

PATENTS

Autonomous robotic cutting system · 18/377,196 · Filed Oct 5, 2023
Feature-driven next view planning of 3-dimensional surfaces · 18/119,547 · Filed Mar 9, 2023
Salvage metal cutting robot · 17/721,553 · Filed Apr 15, 2022

Updated: November 2024